

TUTORIAL

Mastering a simulation-based parametric method to obtain bias-corrected point estimates and sampling variance for effect sizes

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Abstract

Meta-analyses require an effect-size estimate and its corresponding sampling variance from primary studies. For some effect size statistics, however, estimators of sampling variance are unavailable, requiring new derivations. Traditionally, such formulas are obtained using hand-derived Taylor expansions (the delta method), but this approach can be technically demanding and inaccessible to many applied researchers. Building on the idea of single-fit parametric resampling, we introduce the SAFE bootstrap: a Single-fit, Accurate, Fast, and Easy simulation recipe that replaces complex algebra with four intuitive steps: fit, draw, transform, and summarise. SAFE is a model-based parametric bootstrap applied to summary statistics, so its performance depends on how well the assumed sampling model approximates the sampling distribution of the reported statistics. Here, we focus on two-group effect sizes that can be parameterised from standard reported summaries, using Gaussian working models for continuous outcomes and binomial or multinomial models for discrete outcomes. Within this framework, SAFE yields model-based, bias-corrected point estimates and standard errors for familiar effect sizes, and readily extends to less common statistics. We first demonstrate SAFE with a simple example, then apply it to common effect sizes, such as the standardised mean difference and the log odds ratio, as well as several less common measures. With additional coding, SAFE can also accommodate zero values and small sample sizes, albeit with important caveats. Our tutorial, accompanied by R code supplements, aims both to clarify the logic of simulation-based estimation from summary statistics and to provide a practical workflow when only summary statistics are available.

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Highlights**What is already known?**

- Sampling variances and small-sample bias corrections for effect sizes are usually obtained via delta-method expansions, which are hard to derive for non-statisticians.
- Parametric bootstrapping and ‘single-fit’ resampling have been proposed for inference, but their use for obtaining bias-corrected (BC) effect-size estimates and SEs in meta-analysis remains limited.

What is new?

- We formalise four steps (fit, draw, transform, and summarise) for the Single-fit, Accurate, Fast and Easy (SAFE) bootstrapping procedure, to obtain model-based BC point estimates and standard errors from reported summary statistics.
- We illustrate SAFE for common two-group designs using Gaussian working models for continuous outcomes and binomial (multinomial) models for discrete outcomes, including paired designs.

Potential impact for RSM readers

- We lower the technical barrier to introducing or adopting novel effect sizes and enable routine bias correction without new algebraic formulas.
- We provide a reproducible workflow that complements formula-based methods and makes their assumptions more transparent with code and an additional material (<https://ejlundgren.github.io/SAFE/>).

1. Introduction

More than four decades ago, statisticians realised that, instead of deriving sampling distributions with pencil-and-paper algebra, they could let the computer do the heavy lifting. With modern computing, one can repeatedly generate pseudo-data, recalculate the statistic of interest, and use the resulting distribution for inference. This broad family of approaches, often grouped under ‘bootstrapping’, has become influential in many disciplines, including meta-analysis.^{1–4} However, not all bootstrap procedures are alike: some resample raw data non-parametrically, whereas others simulate from an explicitly assumed parametric model. Our focus here is on the latter.

Of relevance, formulas for common effect size statistics (e.g., standardised mean difference [SMD]) and their sampling variances are frequently used in meta-analysis. These formulas are often obtained via first- or second-order Taylor expansions, also known as the ‘delta method’.⁵ Such formulas can be elegant and useful, but they may be difficult to derive for new statistics, difficult to verify, and sometimes inaccurate in small samples or under strong skewness. For many practitioners, the main practical problem is therefore not only how to derive a formula, but how to know whether the resulting approximation is adequate.

In this tutorial, we show how a parametric technique, termed Single-fit, Accurate, Fast, and Easy-to-implement (SAFE) bootstrapping, can provide model-based bias corrections and model-based standard errors (SEs) without requiring a new analytic derivation for every effect size. SAFE is not a non-parametric (assumption-free) bootstrap.^{1–4} Rather, it is a bootstrap on summary statistics, conditional on an explicitly chosen sampling model. That is, SAFE works by fitting a simple sampling model once to the reported statistics, drawing many replicates under that model, transforming each replicate into the effect size of interest, and summarising the resulting bootstrap cloud. Its performance, therefore, depends on how well the assumed sampling model approximates the sampling distribution of the reported statistics.

The idea explored in this tutorial is not new, and previous papers have hinted at the power of this single-fit approach.^{6,7} Our contribution is to organise it into a clear four-step workflow for meta-analytic effect sizes and to demonstrate it in settings commonly encountered by meta-analysts. Specifically, we focus on two-group designs that can be parameterised from standard reported summaries: Gaussian working models for continuous outcomes and binomial or multinomial models for discrete outcomes.

We also emphasise that this tutorial mainly concerns effect sizes on transformed scales that are approximately unbounded and reasonably symmetric (e.g., log response ratio, log odds ratio [OR] and log risk ratio [RR]), because these are the settings in which the working models used here are most natural.

We begin with a simple example to make the SAFE bootstrap recipe accessible and concrete. Before moving to specific effect sizes, we also provide a brief practical guide summarising the designs covered here, the required summary statistics, and the working sampling models assumed. We then work through continuous-outcome effect sizes, namely the log response ratio and the SMD, including paired designs, before turning to discrete outcomes, such as log ORs and log RRs. We also illustrate less common metrics, such as the log coefficient-of-variation ratio^{8,9} and the Hardy–Weinberg disequilibrium (HWD) statistic.¹⁰ Throughout, we compare SAFE with familiar formula-based estimators and clarify when SAFE should be viewed as a useful practical alternative. An online supplement (<https://ejlundgren.github.io/SAFE/>) provides additional R code¹¹ and simulation results. Together, we hope that medical, biological, and social scientists, along with their students, can feel confident and ready to utilise this underappreciated yet broadly applicable method, which may encourage researchers to employ other simulation-based approaches more generally.

2. The concept and a simple example

2.1. Four steps to SAFETy

The SAFE bootstrap procedure is best understood in terms of four steps. Throughout, let θ denote the effect-size statistic of interest. Importantly, every step below is conditional on an assumed sampling model for the reported summary statistics:

1. *Fit*: Choose a plausible sampling model for the observed data or summary statistics (e.g., a normal/Gaussian model for means or a Binomial model for counts) and estimate its parameters from the observed study. This step occurs only once and is therefore considered a ‘single-fit’ procedure. SAFE is therefore a parametric, model-based bootstrap on reported summary statistics, not a non-parametric bootstrap on raw data. Its validity depends on whether this working model provides a reasonable approximation to the sampling distribution of the statistics being used.
2. *Draw*: Generate a large batch of simulated summary statistics directly from the fitted model in one command (e.g., `rnorm()` or `rbinom()`). Under the assumed sampling model, each draw can be viewed as a credible representation of what might have been observed had the study been repeated under the same conditions.
3. *Transform*: Apply the effect-size formula to every simulated data set. Because transformations can be vectorised in software such as R, we turn tens of thousands of draws into tens of thousands of effect sizes very quickly.
4. *Summarise*: From the resulting vector of effect size estimates $\{\theta^*\}$ (or $\theta^* = \theta_b^*$, for $b = 1, \dots, B$; the vector $\{\theta^*\}$ sometimes referred to as a bootstrap cloud), we first quantify the standard deviation of the vector of effect size estimates, which is an estimate of the SE for the parameter, denoted as SE_{SAFE} :

$$SE_{SAFE} = sd(\theta^*). \quad (1)$$

Alternatively, we can calculate sampling variance as

$$Var_{SAFE} = var(\theta^*). \quad (2)$$

Then, we can quantify bias in the original effect size estimate ($\hat{\theta}$; the hat indicates that it is an estimate or estimator, an attempt to estimate the true values θ , an estimand; note that, for example, Cohen’s d and Hedges’ g are estimators of the standardised mean difference) by subtracting the

original estimate from the mean of the vector of effect size estimates ($\bar{\theta}^*$), simulated from Step 2:

$$\widehat{\text{bias}} = \bar{\theta}^* - \hat{\theta}. \quad (3)$$

Using this estimated bias (an explanation of this formula is given below), we can obtain the bias-corrected (BC) point estimate, denoted as $\hat{\theta}_{\text{BC}}$:

$$\hat{\theta}_{\text{BC}} = \hat{\theta} - \widehat{\text{bias}}. \quad (4)$$

By combining Eqs. (3) and (4), we obtain

$$\hat{\theta}_{\text{BC}} = \hat{\theta} - (\bar{\theta}^* - \hat{\theta}) = 2\hat{\theta} - \bar{\theta}^*. \quad (5)$$

These four steps are visualised in Figure 1 using a simple example presented in the next section.

The bias correction via SAFE is elegant in its simplicity.^{3,12} Before understanding how this bias correction works, it is essential to comprehend how such a bias arises. Most meta-analytic effect sizes are non-linear transformations of the original (raw) statistics (e.g., mean and SD). When a transformation function $f(\cdot)$ is curved, the expected value of a transformed random variable (x) is not equal to the transformation of the variable's expected value (also known as Jensen's inequality):

$$E[f(x)] \neq f(E[x]). \quad (6)$$

Previous bias corrections for the log response ratio are essentially algebraic approximations to the same curvature-induced bias that SAFE estimates directly by simulation under the fitted sampling model.¹³ By subtracting the estimated bias, we immediately obtain a second-order Taylor expansion correction that is often equivalent but without using the delta method. Importantly, when sample sizes are small, or uncertainty is large, this transformation-induced curvature bias can become non-negligible (Figure 1). SAFE is designed to estimate that bias directly, but only conditional on the chosen sampling model; if that model is misspecified, the estimated bias may also be misleading.

The key point is that SAFE estimates the bias of the plug-in estimator under the fitted sampling model. Let $\hat{\theta}$ denote the effect-size estimate computed from the observed summary statistics, and let $\theta_1^*, \dots, \theta_B^*$ denote the corresponding bootstrap replicates obtained after drawing from the fitted model and reapplying the same effect-size formula. The average bootstrap replicate, $\bar{\theta}^* = \frac{1}{B} \sum_{b=1}^B \theta_b^*$, approximates the expected value of the estimator under the fitted model, $E_{\text{model}}(\hat{\theta})$. Given this and the definition, $\text{bias} = E[\hat{\theta}] - \theta$, the bootstrap estimate of bias is $\widehat{\text{bias}}_{\text{SAFE}} = \bar{\theta}^* - \hat{\theta}$, which estimates the difference between the model-based expectation of the estimator and the observed plug-in estimate. Subtracting this estimated bias gives the usual bootstrap BC estimator: $\hat{\theta}_{\text{BC}} = \hat{\theta} - \widehat{\text{bias}}_{\text{SAFE}} = 2\hat{\theta} - \bar{\theta}^*$. Because this is a model-based bias correction, it does not directly estimate bias relative to an unknown true parameter; rather, it estimates the bias of the plug-in estimator under the assumed sampling model. Relatedly, SAFE is conditional on the fitted sampling model so that terms such as 'accurate', 'bias-corrected', and 'standard error' should be read throughout this tutorial as model-based. When the working model is poor, SAFE can also perform poorly; in that sense, SAFE complements rather than eliminates the need for statistical judgement.

Before presenting this example, we briefly note why our approach earns the acronym SAFE: Single-fit, Accurate, Fast, and Easy(-to-implement). It is because the procedure comprises: (i) *single-fit* with a sampling model using the data; (ii) *accuracy* through a bias-correction procedure; (iii) *fast* estimation as one vectorised simulation replaces thousands of re-fits; and (iv) an *easy* method, which, as we show below, can use the same four steps for any effect-size statistic. We return to each of these components as we work through our examples.

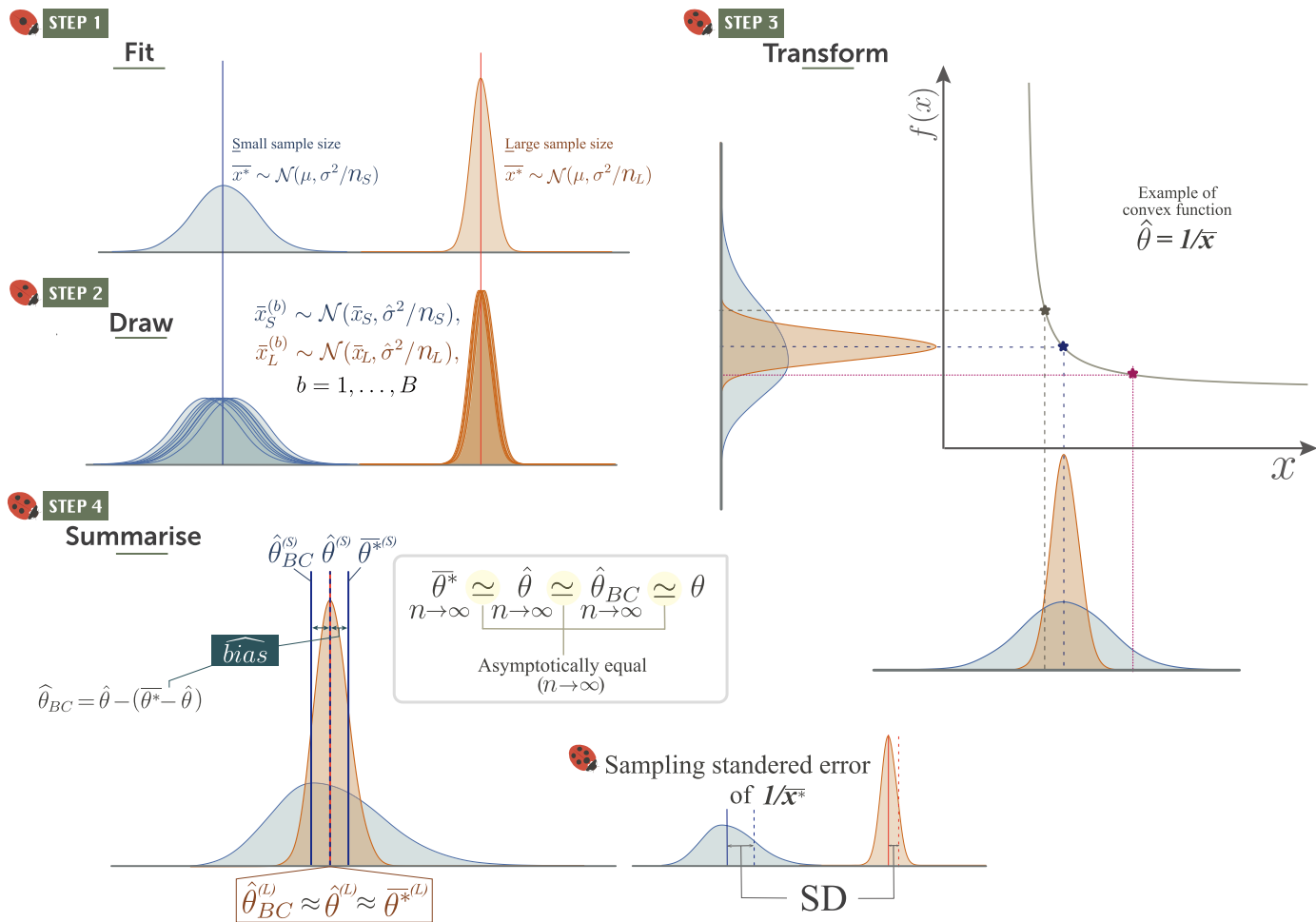


Figure 1. Visualising four steps of the SAFE bootstrapping, using an example with a convex function, $1/x$. S signifies a small sample size case while L a large sample size case; this schematics visualises why Jensen's inequality has little impact on the large sample size case (for other symbols, see the main text). For why and how small-sample bias correction works, see also Figure 2.

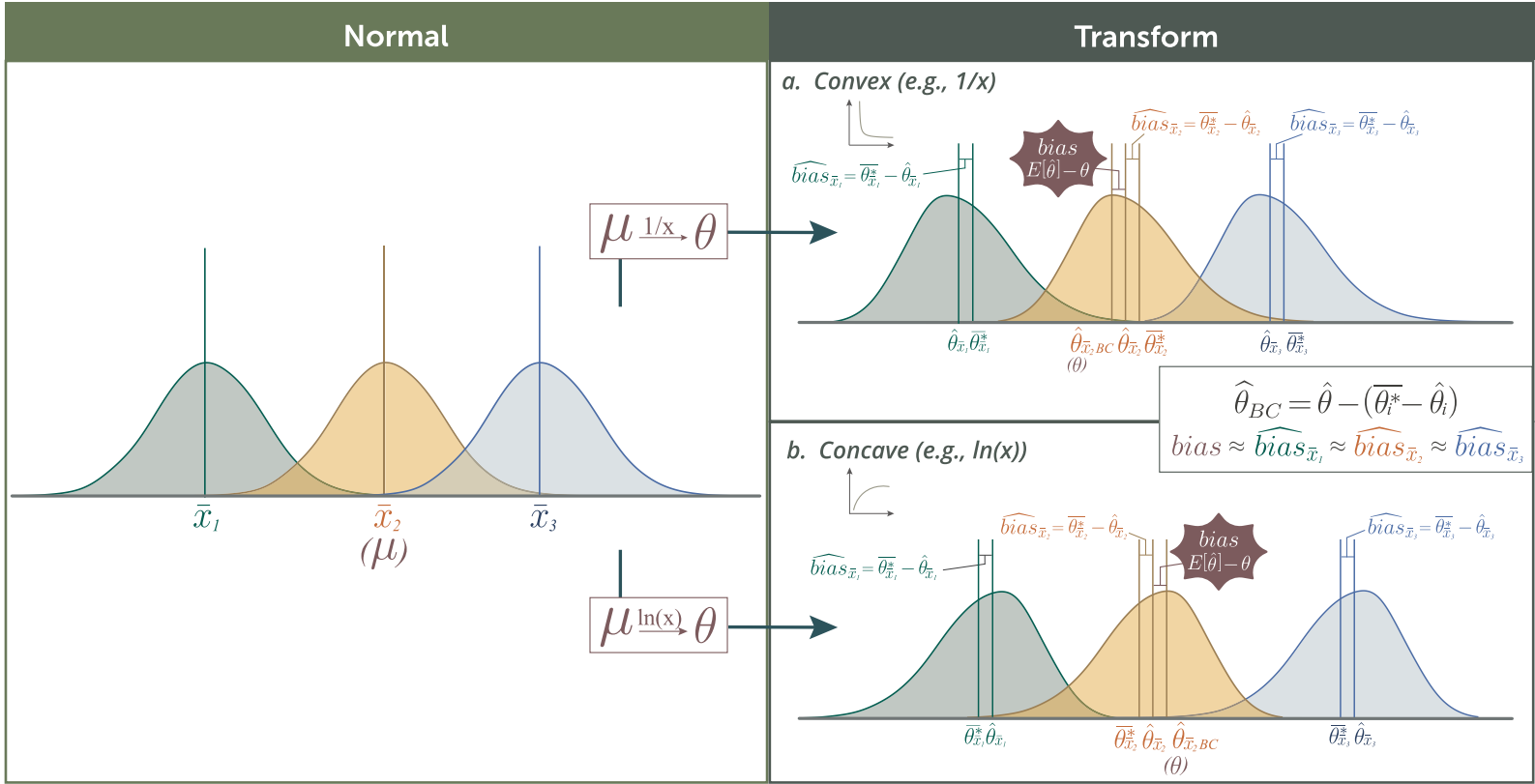


Figure 2. Bootstrap bias correction under convex versus concave transforms. Left panel: Three possible sampling distributions of the sample mean $\bar{x}_1, \bar{x}_2, \bar{x}_3 \sim N(\mu_i, \sigma^2/n)$ with identical spread (same SE) but different locations. Right panel: Applying (a) a convex transform (e.g., $f(x) = 1/x$) or (b) a concave transform (e.g., $f(x) = \ln x$) bends these distributions. By Jensen's inequality, $E[f(\bar{x})] \geq f(E[\bar{x}])$ with ' $\geq$ ' for convex and ' $\leq$ ' for concave, yielding an upward (a) or downward (b) bias. For each location i , we simulate $\{\bar{x}_i^*\}$, transform to $\bar{\theta}_i^* = f(\bar{x}_i^*)$, and use $\bar{\theta}_i^* \approx E[\hat{\theta}_i]$ to estimate bias: $\widehat{bias}_i = \bar{\theta}_i^* - \hat{\theta}_i$. The bias-corrected estimator is $\hat{\theta}_{BC} = \hat{\theta} - \widehat{bias}_i$. Because the three $\bar{x}_i$ distributions have the same SE, the bias (the distance between $\bar{\theta}_i^*$ and $\hat{\theta}_i$) is essentially the same across i , even though the $\bar{x}_i$ (and thus $\hat{\theta}_i = f(\bar{x}_i)$) differ in location. In short, the bootstrap correction is driven by curvature and uncertainty, not by where $\bar{x}$ happens to sit.

2.2. A simple example

Imagine an experiment to measure an insect's performance in solving a maze (in seconds, denoted as x) with five insects. The mean completion time, $\bar{x}$, is a measure of the latency to finish a maze, so the longer it takes, the worse the performance is. To make the readout more intuitive, we may wish to take the reciprocal of the mean, so larger values mean a faster time to completion (e.g., the reciprocal of 10 sec is 0.1 while 5 sec is 0.2). This reciprocal can then be considered as a measure of 'speed'. Meta-analytically, we may wish to use this reciprocal, R , as our new effect size statistic:

$$R = \frac{1}{\bar{x}}, \quad (7)$$

where $\bar{x}$ is the average of the time taken for maze completion for five insects. Based on this transformation, it is not immediately apparent how to calculate the sampling variance, or SE, for R (speed). Further, because the reciprocal is a curved (convex) function, the plug-in estimate $1/\bar{x}$ is known to be upwardly-biased (i.e., $1/\bar{x} > \text{mean}(1/x)$) and especially so in small samples (i.e., Jensen's inequality). Now we apply our four-step SAFE procedure to R ($1/x$) (Figure 1).

Step 1: Fit once. The sample mean ($\bar{x}$) has the following sampling distribution exactly when the underlying observations are normally distributed, and approximately so more generally by the central limit theorem when the observations are non-normal but the sample size is reasonably large:

$$\bar{x}^* \sim \mathcal{N}(\mu, \sigma^2/n),$$

where $\bar{x}^*$ indicates an instance of $\bar{x}$, μ is the population mean, σ^2 is the population variance, and n is the sample size. Using the observed sample, we plug in $\mu \approx \bar{x}$ and $\sigma \approx \text{sd}(x)$, so that the working model for the sampling distribution of the mean becomes $\bar{x}^* \sim \mathcal{N}(\bar{x}, \text{sd}(x)^2/n)$. Note that the standard deviation of $\bar{x}^*$ is $\hat{\sigma}/\sqrt{n}$, not $\hat{\sigma}$; the latter being the population standard deviation and the former equivalent to the SE (i.e., the standard deviation of a sampled statistic). As mentioned above, SAFE deals with a distribution of parameters (e.g., mean), not raw data, which tend to be inaccessible to the meta-analyst.

Step 2: Draw once. We draw B surrogate sample means from this normal approximation. Larger values of B are better but computationally costly. We recommend $B = 100,000$, although $B = 1,000,000$ could slightly improve performance, as validated in our simulation (<https://ejlundgren.github.io/SAFE/>), and offers an excellent balance between accuracy and efficiency.

Step 3: Transform once. Each surrogate mean is converted to R (a speed) by taking its reciprocal.

Step 4: Summarise two metrics. We obtain BC point and SE estimates from bootstrapped data as follows:

$$\text{SE}_{\text{SAFE}} = \text{sd}(R^*) \quad \text{and} \quad R_{\text{BC}} = 2\hat{R} - \overline{R^*}.$$

The entire procedure is only a few lines of R code; note that crucially $\text{sd}(x)\sqrt{n}$ is used in the call to `rnorm()`, ensuring that we simulate data that follow the distribution of sample means:

```
set.seed(1)
x <- c(11.3, 9.7, 10.4, 12.0, 9.1) # five maze completion times
n <- length(x) # number of observations in vector x
xbar <- mean(x) # calculate mean of observations
Rhat <- 1 / xbar # mean speed

## --- SAFE bootstrap -----
mu <- xbar # mean time taken
sigma <- sd(x) / sqrt(n) # calculate SD of the mean
B <- 1e6 # 1 million bootstrap draws
x_star <- rnorm(B, mu, sigma) # generating possible xbars (mean time taken)
R_star <- 1 / x_star # transform to speed
```

```

bias    <- mean(R_star) - Rhat           # bootstrap bias
R_BC    <- Rhat - bias                   # calculate bias-corrected speed
se_SFB  <- sd(R_star)                   # calculate SAFE standard error

round(c(Rhat = Rhat, R_BC = R_BC, SE_SAFE = se_SFB), 4)

```

The console prints the following:

```

Rhat    R_BC SE_SAFE
0.0952  0.0950 0.0048

```

Three things are notable from this simple example. With a small sample size of five observations, SAFE delivers a bias correction of roughly 2% $0.0952/0.0950 \approx 1.002$; one can immediately see the degree to which such a correction matters. The SE emerges automatically; no delta-method derivation of $\text{Var}(1/\bar{x})$ is required. Nothing in the code is specific to the reciprocal itself. Replace the line `R_star <- 1 / x_star` with another transformation, such as an SMD or a log OR, and the same four-step logic still applies. What changes is the required sampling model for the summary statistics entering that transformation. Accordingly, SAFE yields a BC estimate and an SE under the assumed sampling model, rather than a universally valid SE irrespective of model choice (for simulation validation, see our additional material: <https://ejlundgren.github.io/SAFE/>); incidentally, the delta-method-based variance for R is $\text{var}(R) = s^2/(n\bar{x}^4)$.

Before turning to specific examples, it is useful to state the practical scope of this tutorial. Here, we focus on effect sizes for two-group designs that can be computed from standard reported summaries (means, standard deviations, sample sizes, and cell counts). In the continuous examples, we use Gaussian working models for sample means and, where needed, for sample variances or related quantities. In the discrete examples, we use binomial or multinomial working models for counts.

3. Effect sizes for continuous outcomes

In this section, we show how the same four-step SAFE logic extends to effect-size statistics for continuous outcomes comparing two groups, namely the log response ratio (ratio of means) and the standardised mean difference. First, we consider two independent groups, and then two dependent groups (paired, matched, or crossover design).

3.1. Response ratio (ratio of means: lnRoM)

The so-called response ratio (log response ratio; also known as the log ratio of means) was first proposed by Hedges et al.¹⁴ as an alternative to the SMD, which we denote as lnRoM. We note that the acronym lnRR is commonly used, but in this article, we reserve it for the log risk ratio presented in the next section. Imagine two groups, Group 1 ($\bar{x}_1$, s_1 , n_1 ; sample mean, standard deviation, and sample size) and Group 2 ($\bar{x}_2$, s_2 , n_2). The original log response ratio can be written as

$$\ln \text{RoM}_1 = \ln \left(\frac{\bar{x}_1}{\bar{x}_2} \right). \quad (8)$$

This is the definition formula for lnRoM; such formulas are often referred to as ‘plug-ins’. A first-order Taylor expansion gives the sampling variance as

$$\text{Var}[\ln \text{RoM}_1] = \frac{s_1^2}{n_1 \bar{x}_1^2} + \frac{s_2^2}{n_2 \bar{x}_2^2}. \quad (9)$$

Note that we label this combination of Eqs. (8) and (9) as ‘First’ in our R scripts and this tutorial, although Eq. (9) was not derived using the delta method and the first-order Taylor expansion. Later

Lajeunesse¹³ improved both point and variance estimates by using second-order Taylor expansions (which we label as ‘Second’), which are as follows:

$$\ln \text{RoM}_2 = \ln \left(\frac{\bar{x}_1}{\bar{x}_2} \right) + \frac{1}{2} \left(\frac{s_1^2}{n_1 \bar{x}_1^2} - \frac{s_2^2}{n_2 \bar{x}_2^2} \right). \quad (10)$$

$$\text{Var}[\ln \text{RoM}_2] = \frac{s_1^2}{n_1 \bar{x}_1^2} + \frac{s_2^2}{n_2 \bar{x}_2^2} + \frac{1}{2} \left(\frac{s_1^4}{n_1^2 \bar{x}_1^4} + \frac{s_2^4}{n_2^2 \bar{x}_2^4} \right). \quad (11)$$

For SAFE bootstrapping, we treat the pair of sample means as a bivariate normal distribution:

$$\begin{bmatrix} \bar{x}_1^* \\ \bar{x}_2^* \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix}, \begin{bmatrix} s_1^2/n_1 & 0 \\ 0 & s_2^2/n_2 \end{bmatrix} \right).$$

At this stage, one could use two independent normal distributions (i.e., `rnorm()` twice). However, the multivariate normal distribution has an advantage, which we will show later. A single call to `mvrnorm()` from the MASS package¹⁵ draws B pairs $(\bar{x}_1^*, \bar{x}_2^*)$; we transform each into $\ln(\bar{x}_1^*/\bar{x}_2^*)$ in one vectorised step, then compute

$$\text{SE}_{\text{SAFE}} = \text{sd} \left(\ln \left(\frac{\bar{x}_1^*}{\bar{x}_2^*} \right) \right), \quad \text{and} \quad \ln \text{RoM}_{\text{BC}} = 2 \ln \left(\frac{\bar{x}_1}{\bar{x}_2} \right) - \overline{\ln \left(\frac{\bar{x}_1^*}{\bar{x}_2^*} \right)}.$$

Under this Gaussian working model, simulated means may occasionally fall below zero, which is incompatible with the log transformation. In practice, however, this issue is typically negligible when the observed means are largely positive relative to their sampling variation, making negative draws rare. More importantly, the multivariate normal formulation is convenient because it generalises naturally to settings where the two sample means are correlated, such as paired or otherwise dependent designs. We therefore use `mvrnorm()` as the default working model throughout, while recognising that its suitability depends on the probability of implausible negative draws being very small (note that we need to remove such negative values for the SAFE procedures).

Below is an example R code in which we compare SAFE estimates with estimates from the formulas in Eqs. (8) through (11). Estimates from the second-order delta method are generally considered more accurate than those from the first-order counterpart, and we expect the SAFE estimates to be similar to the second-order estimates.

```
set.seed(7)
## Example data (means, SDs, sample sizes) -----
x1 <- 13.4; s1 <- 4.6; n1 <- 18          # group 1
x2 <- 16.1; s2 <- 3.9; n2 <- 17          # group 2

## 1. FIRST-order (plug-in) lnRoM & SE -----
theta_First <- log(x1 / x2)              # Eq. 8
var_First   <- s1^2/(n1 * x1^2) + s2^2/(n2 * x2^2) # Eq. 9
se_First    <- sqrt(var_First)

## 2. SECOND-order lnRoM & SE -----
theta_Second <- log(x1 / x2) +
  0.5 * ( s1^2/(n1 * x1^2) - s2^2/(n2 * x2^2) ) # Eq. 10

var_Second <- var_First +
  0.5 * ( s1^4/(n1^2 * x1^4) + s2^4/(n2^2 * x2^4) ) # Eq. 11
```

```

se_Second <- sqrt(var_Second)

## 3. SAFE single-fit bootstrap (multivariate Normal) -----
library(MASS)
mu <- c(m1 = x1, m2 = x2)
Sig <- diag(c(s1^2/n1, s2^2/n2))          # independent sample means
B <- 1e6                                  # number of bootstrap draws
draw <- mvrnorm(B, mu, Sig)               # Fit-once, draw-once
theta_star <- log(draw[, "m1"] / draw[, "m2"]) # Transform-once

se_SAFE <- sd(theta_star)                 # first summary: SE
bias_SAFE <- mean(theta_star) - theta_First
theta_SAFE_BC <- theta_First - bias_SAFE   # second summary: BC point

## 4. Print tidy comparison -----
results <- rbind( Point = c(First = theta_First,
                           Second = theta_Second,
                           SAFE = theta_SAFE_BC),
  SE = c(First = se_First,
         Second = se_Second,
         SAFE = se_SAFE)
)
round(results, 4)

```

When you run the code, the console returns something like the following, where *First* represents estimates from Eqs. (8) and (9), *Second* represents estimates from Eqs. (10) and (11), and *SAFE* represents estimates from the SAFE bootstrapping.

	First	Second	SAFE
Point	-0.1836	-0.1820	-0.1820
SE	0.1000	0.1001	0.1007

As you see, the point estimates from the second column (the second-order correction formulas; Eqs. (10) and (11)) and the third column (SAFE bootstrap) match well, although SE from SAFE is slightly larger. In contrast, the point estimate from the first column (Eqs. (8) and (9)) deviates from the other two. Note that SE_{SAFE} can be slightly larger in small samples, which usually results in higher coverages (the coverages are presented in an additional material: <https://ejlundgren.github.io/SAFE/>).

3.2. Standardised mean difference (SMD)

The SMD remains extremely popular for continuous outcomes in ecology, psychology, and medicine. SMD may also be often referred to as Cohen's d or Hedges' g , both of which are estimators of SMD. Although Cohen seems to have introduced SMD first,¹⁶ Hedges¹⁷ was the first to formalise SMD with its sampling variance. As with InRoM, we have two independent groups with summary statistics: Group 1 ($\bar{x}_1, s_1, n_1$) and Group 2 ($\bar{x}_2, s_2, n_2$). Following Borenstein et al.,¹⁸ we can define Cohen's d as

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_p}, \quad s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}, \quad (12)$$

where the term $n_1 + n_2 - 2$ is often referred to as the degree of freedom (df). The sampling variance for SMD is

$$\text{Var}[d] = \frac{n_1 + n_2}{n_1 n_2} + \frac{d^2}{2(n_1 + n_2 - 2)}. \quad (13)$$

We note that this variance formula can be traced back to Hedges,¹⁷ who notably used the non-central t distribution in its derivation, rather than the delta method, which is commonly used elsewhere. Hedges¹⁷ recognised that, for small samples, the expectation of d is slightly inflated.^{19,20} Multiplying by the correction factor:

$$J = 1 - \frac{3}{4(n_1 + n_2 - 2) - 1}, \quad (14)$$

can alleviate this inflation; the resulting so-called Hedges' g can be written as

$$g = J \left(\frac{\bar{x}_1 - \bar{x}_2}{s_p} \right). \quad (15)$$

And, therefore, the sampling variance of g is

$$\text{Var}[g] = J^2 \text{Var}[d] = J^2 \left(\frac{n_1 + n_2}{n_1 n_2} + \frac{d^2}{2(n_1 + n_2 - 2)} \right). \quad (16)$$

Given that SMD includes the sample standard deviation (s_p or variance s_p^2), we need to know the sampling variance of the variance estimate, which, assuming variances are normally distributed, is²¹:

$$\text{Var}(\sigma^2) = \frac{2\sigma^4}{n - 1}. \quad (17)$$

We note that variance is bounded at zero, and so when the variance is close to zero, the assumption of normality is impractical. For such cases, we discuss an alternate estimator below.

We also note that the SAFE bootstrap of SMD is by far the most challenging among those presented in this article. Now, for the SAFE procedure, we sample the quartet $(\bar{X}_1, \bar{X}_2, S_1^2, S_2^2)$ jointly:

$$\begin{bmatrix} \bar{x}_1^* \\ \bar{x}_2^* \\ s_1^{2*} \\ s_2^{2*} \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \\ s_1^2 \\ s_2^2 \end{bmatrix}, \underbrace{\begin{bmatrix} \frac{s_1^2}{n_1} & 0 & 0 & 0 \\ 0 & \frac{s_2^2}{n_2} & 0 & 0 \\ 0 & 0 & \frac{2s_1^4}{n_1 - 1} & 0 \\ 0 & 0 & 0 & \frac{2s_2^4}{n_2 - 1} \end{bmatrix}}_{\Sigma} \right).$$

Off-diagonal elements of Σ are zero because the two groups are assumed independent. In addition, for independently and identically distributed (i.i.d.) normal data, the sample mean, and sample variance are independent, which motivates the block-diagonal form used here. As before, using `mvrnorm()`, we draw B independent replicates of the quartet: $(\bar{x}_1^*, \bar{x}_2^*, s_1^{2*}, s_2^{2*})$. Then, we vectorise the usual pooled-variance formula and compute d using Eq. (12). A more principled alternative is to sample variance components from their exact or more natural positive-support distributions, such as χ^2 distributions (or the Wishart distribution in multivariate settings), rather than from a Gaussian approximation. These approaches avoid impossible negative variance draws and are likely preferable when implementing SAFE for statistics that depend directly on sample variances. For simplicity, however, we retain the Gaussian working model here and return to this issue in an additional material (<https://ejlundgren.github.io/SAFE/>).

From this vector of d^* values, we read off:

$$SE_{SAFE} = sd(d^*) \quad \text{and} \quad d_{BC} = 2\hat{d} - \overline{d^*}.$$

Below is an R example where we implement formulas for d and g along with the SAFE procedure.

```
set.seed(11)
## 0. Example data -----
x1 <- 13.4; s1 <- 4.6; n1 <- 18 # group 1
x2 <- 16.1; s2 <- 3.9; n2 <- 17 # group 2
df <- n1 + n2 - 2                # degrees of freedom required for sp2

## 1. Plug-in Cohen d & delta-method SE -----
sp2 <- ((n1-1)*s1^2 + (n2-1)*s2^2) / df
d_hat <- (x1 - x2) / sqrt(sp2)    # Eq. 12
Var_d <- (n1+n2)/(n1*n2) + d_hat^2/(2*df) # Eq. 13
se_delta <- sqrt(Var_d)

## 2. Hedges' g & its SE -----
J <- 1 - 3/(4*df - 1)
g_hat <- J * d_hat                # Eq. 15
Var_g <- J^2 * Var_d              # Eq. 16
se_g <- sqrt(Var_g)

## 3. SAFE bootstrap on d (4-variate) -----
library(MASS)
mu <- c(X1=x1, X2=x2, V1=s1^2, V2=s2^2)
Sigma <- matrix(0, 4, 4)
diag(Sigma) <- c(s1^2/n1, s2^2/n2,
                 2*s1^4/(n1-1), 2*s2^4/(n2-1))
B <- 1e6 # the number of bootstrap
draw <- mvrnorm(B, mu, Sigma)

# reject negative variances
draw <- draw[draw[, "V1"]>0 & draw[, "V2"]>0, ]

sp2_star <- ((n1-1)*draw[, "V1"] + (n2-1)*draw[, "V2"]) / df
d_star <- (draw[, "X1"] - draw[, "X2"]) / sqrt(sp2_star)

se_SAFE <- sd(d_star)
bias <- mean(d_star) - d_hat
d_BC <- d_hat - bias

## 4. Compare -----
res <- rbind( Point = c(d = d_hat, g = g_hat, SAFE = d_BC),
              SE = c(d = se_delta, g = se_g, SAFE = se_SAFE)
            )
round(res, 4)
```

We get the following in the console:

```
      d      g    SAFE
Point -0.6316 -0.6171 -0.6156
SE      0.3470  0.3391  0.3613
```

We observe that the BC d via SAFE is closer to g than to the uncorrected d in this example. Also, SE_{SAFE} is somewhat larger here than the formula-based alternatives. Again, this should not

automatically be interpreted as undue conservatism: simulation-based SEs can reflect higher-order uncertainty induced by the non-linear transformation, especially in smaller samples. A full simulation comparison is given in an additional material (<https://ejlundgren.github.io/SAFE/>); in that simulation, we see that the use of Wishart distributions produces better point and variance estimators than an assumed normal distribution. We also find the SE for d (Eq. (13)) seems to perform best in combination with Hedge's g , and notably, `escalc()` from the `metafor` package²² uses this combination.

3.3. Matched (paired) designs or repeated measures

Many experiments measure the same individuals twice: before/after a treatment, or under two conditions. These designs are known to be more powerful because the SE can be smaller than an independent design. This is because the within-pair correlation r can reduce the sampling variance. This reduction can be seen in action in the formula for the sampling variance for InRoM for dependent cases (denoted by the subscript 'D')²³:

$$\text{Var}[\text{In RoM}_D] = \frac{s_1^2}{n_1 \bar{x}_1^2} + \frac{s_2^2}{n_2 \bar{x}_2^2} - \frac{2 r s_1 s_2}{\bar{x}_1 \bar{x}_2 \sqrt{n_1 n_2}}. \quad (18)$$

Note that the within-pair correlation r is ideally estimated from the original study, but is often not reported. In such cases, analysts may need to assume a plausible value (e.g., $r = 0.5$).²⁴ However, because both formula-based and SAFE estimates can depend on this assumption, we recommend conducting sensitivity analyses over a range of plausible r values and reporting how the point estimate and sampling variance change across that range along how model estimates change. The formula for the point estimate for dependent InRoM is as above (Eq. (8)). Also, with a paired design, $n_1 = n_2 = n$.

To estimate the bias-adjusted point estimates and sampling variances for dependent InRoM via SAFE, we simply need to adjust the assumed distribution of the sample statistics in the 'fit' step to incorporate the correlation (as covariance) into the bivariate normal distribution from which we draw our samples as

$$\begin{bmatrix} \bar{x}_1^* \\ \bar{x}_2^* \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix}, \frac{1}{n} \begin{bmatrix} s_1^2 & r s_1 s_2 \\ r s_1 s_2 & s_2^2 \end{bmatrix}\right).$$

From here, the remaining steps for SAFE estimation are identical to the independent case of InRoM.

Similarly, there are sampling variance formulas for dependent SMD cases.^{25,26} However, again, we can use SAFE for estimation by specifying the joint sampling distribution of the two group means and their sample variances by a quartet-variate normal with covariances.⁹ A slight complication is that we need to know the sampling covariance of two variances, which is defined as²¹:

$$\text{Cov}(\sigma_1^2, \sigma_2^2) = \frac{2\rho^2 \sigma_1^2 \sigma_2^2}{n-1}, \quad (19)$$

where ρ is the true value for an estimated within-pair correlation r .

In the fit step of SAFE, we then draw once from

$$\begin{bmatrix} \bar{x}_1^* \\ \bar{x}_2^* \\ s_1^{2*} \\ s_2^{2*} \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \\ s_1^2 \\ s_2^2 \end{bmatrix}, \begin{bmatrix} \frac{s_1^2}{n} & \frac{r s_1 s_2}{n} & 0 & 0 \\ \frac{r s_1 s_2}{n} & \frac{s_2^2}{n} & 0 & 0 \\ 0 & 0 & \frac{2 s_1^4}{n-1} & \frac{2 r^2 s_1^2 s_2^2}{n-1} \\ 0 & 0 & \frac{2 r^2 s_1^2 s_2^2}{n-1} & \frac{2 s_2^4}{n-1} \end{bmatrix}\right).$$

Again, the rest of the SAFE procedure remains the same, with the notable exception that $n_1 = n_2 = n$. In summary, for dependent effect sizes, once the correlation r is incorporated into the covariance matrix for the ‘fit’ step, the four SAFE steps remain unchanged.

4. Effect sizes for discrete outcomes

Having shown that SAFE handles continuous-outcome effect sizes, we now move to discrete-outcome effect size statistics, namely ORs and RRs. We again use the same four steps, but the sampling model consists of a pair of independent binomial distributions. We also show how to use a continuity correction (e.g., adding 0.5 to every cell value) when zero counts exist.

4.1. Odds ratio (lnOR)

Many studies report a binary outcome (e.g., survived/died, chose patch A/patch B). For two independent groups, we obtain the 2×2 table of counts, as

	Event	No event	Total
Group 1	a	b	$n_1 = a + b$
Group 2	c	d	$n_2 = c + d$

with sample risks $\hat{p}_1 = a/n_1$ and $\hat{p}_2 = c/n_2$. Using these four cells, we can define the log OR as

$$\ln \text{OR} = \ln \left(\frac{a d}{b c} \right), \quad (20)$$

which is a very popular effect size for this type of data.

Under a first-order Taylor expansion, the sample variance for the lnOR is^{27,28}:

$$\text{Var}[\ln \text{OR}] = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}. \quad (21)$$

However, this formula fails whenever a, b, c , or d equals 0, as we cannot divide by 0. A popular workaround is the ‘continuity correction’²⁹ with

$$(a, b, c, d) \longrightarrow (a + 0.5, b + 0.5, c + 0.5, d + 0.5),$$

applied before both the point estimate and variance are evaluated. In settings with zero cells, this means that the practical target of inference is no longer the uncorrected lnOR, but rather the continuity-corrected log OR estimator. Accordingly, when SAFE is used with this correction, it estimates the bias and sampling variance of that corrected estimator.

As mentioned earlier, for SAFE, we model events in each group as independent draws from two binomial distributions, as

$$a^* \sim \text{Binomial}(n_1, p_1), \quad c^* \sim \text{Binomial}(n_2, p_2),$$

drawing B (bootstrap) replicates (a^*, c^*) . Equivalently, each group can be viewed as a two-category multinomial sample with a fixed row total, so that the non-event counts are determined automatically as $b^* = n_1 - a^*$ and $d^* = n_2 - c^*$. Then, we transform each pair into a continuity-corrected bootstrap OR:

$$\ln \text{OR}_{\text{CC}}^* = \ln \left[\frac{(a^* + \delta_a)(d^* + \delta_d)}{(b^* + \delta_b)(c^* + \delta_c)} \right],$$

where the δ terms are zero unless a continuity correction is invoked for that replicate (e.g., $\delta = 0.5$ when any cell is zero).

The vector of $\ln \text{OR}_{\text{CC}}^*$ gives

$$\text{SE}_{\text{SAFE}} = \text{sd}(\ln \text{OR}_{\text{CC}}^*), \quad \widehat{\ln \text{OR}}_{\text{BC}} = 2 \widehat{\ln \text{OR}}_{\text{CC}} - \overline{\ln \text{OR}_{\text{CC}}^*}.$$

Here, the subscript CC indicates that the same continuity-correction rule used for the observed estimator is also applied within the bootstrap replicates whenever needed. Below is an example R script with a continuity correction of 0.5 to handle zero cells in draws.

```
set.seed(24)

## 0. Observed 2x2 table (a b | c d) -----
a <- 2 ; b <- 20          # 2 / 22 events in Group 1
c <- 10; d <- 12          # 10 / 22 events in Group 2
n1 <- a + b
n2 <- c + d

## 1. Plug-in lnOR (no +0.5 since no zero cells) -----
theta_plug <- log((a * d) / (b * c))      # Eq. 20

## the delta-method SE
var_delta <- 1/a + 1/b + 1/c + 1/d        # Eq. 21
se_delta  <- sqrt(var_delta)

## 2. SAFE single-fit bootstrap (Binomial) -----
B <- 1e6                                # bootstrap no.
a_star <- rbinom(B, n1, a / n1)          # events in Group 1
c_star <- rbinom(B, n2, c / n2)          # events in Group 2
b_star <- n1 - a_star                    # non-events G1
d_star <- n2 - c_star                    # non-events G2

## Apply +0.5 only to rows with ANY zero count -----
needs_cc <- (a_star == 0 | b_star == 0 | c_star == 0 | d_star == 0)
a_star[needs_cc] <- a_star[needs_cc] + 0.5
b_star[needs_cc] <- b_star[needs_cc] + 0.5
c_star[needs_cc] <- c_star[needs_cc] + 0.5
d_star[needs_cc] <- d_star[needs_cc] + 0.5

theta_star <- log((a_star * d_star) / (b_star * c_star))

## 3. SAFE summaries -----
se_SAFE <- sd(theta_star)
bias_hat <- mean(theta_star) - theta_plug
theta_BC <- theta_plug - bias_hat        # bias-corrected lnOR

## 4. Nicely formatted output -----
out <- rbind( Point = c(First = theta_plug,
                      SAFE   = theta_BC),
             SE   = c(First = se_delta,
                      SAFE   = se_SAFE)
)
print( round(out, 4) )
```

Running this script should yield the following results:

```
      First  SAFE
Point -2.1203 -1.9515
SE     0.8563  0.8714
```

Both procedures provide comparable point and variance estimates, and their discrepancies disappear as sample sizes increase (see supporting simulation).

4.2. Relative risk (risk ratio: $\ln RR$)

For discrete outcomes, many authors also use the relative risk, also known as the RR. Using the 2×2 contingency table above, the log RR can be defined as

$$\ln RR = \ln\left(\frac{a/n_1}{c/n_2}\right) = \ln\left(\frac{p_1}{p_2}\right). \quad (22)$$

A first-order Taylor expansion for the sampling variance of the observed $\ln RR$,^{28,30} where $p_1 = a/n_1$ and $p_2 = c/n_2$, is

$$\text{Var}[\ln RR] = \frac{1 - p_1}{a} + \frac{1 - p_2}{c}. \quad (23)$$

This expression fails whenever $a = 0$ or $c = 0$. In practice, a continuity correction is then often applied by adding 0.5 to the event and non-event counts within the affected group before evaluating the point estimate and variance. Consequently, when zero-event cells are present, the practical target of inference is the continuity-corrected log RR estimator, rather than the uncorrected $\ln RR$ itself. Accordingly, when SAFE is implemented with this correction, it estimates the bias and sampling variance of that corrected estimator.

For the SAFE procedure, a single call to `rbinom()` for each group suffices just as $\ln OR$ above:

$$a^* \sim \text{Binomial}(n_1, p_1), \quad c^* \sim \text{Binomial}(n_2, p_2),$$

with $b^* = n_1 - a^*$ and $d^* = n_2 - c^*$ filled in by definition. The four SAFE moves: fit, draw, transform, and summarise, then yield:

$$SE_{\text{SAFE}} = \text{sd}(\ln RR_{\text{CC}}^*), \quad \widehat{\ln RR}_{\text{BC}} = 2 \widehat{\ln RR}_{\text{CC}} - \overline{\ln RR_{\text{CC}}^*}.$$

As before, the subscript CC indicates that the same continuity-correction rule used for the observed estimator is also applied within the bootstrap replicates whenever needed. Below is an R example for $\ln RR$ with continuity correction of 0.5, comparing formula-based and simulation-based methods.

```
set.seed(24)

## 0) Observed 2x2 table (a b | c d) -----
a <- 2 ; b <- 20          # 2 / 22 events in Group 1
c <- 10; d <- 12         # 10 / 22 events in Group 2
n1 <- a + b
n2 <- c + d
p1 <- a / n1
p2 <- c / n2

## 1) FIRST-order (plug-in) lnRR & delta SE -----
theta_First <- log(p1 / p2)          # Eq. 22
var_First   <- (1 - p1) / a + (1 - p2) / c  # Eq. 23
se_First    <- sqrt(var_First)

## 2) SAFE single-fit bootstrap (Binomial) -----
B <- 1e6
a_star <- rbinom(B, n1, p1)
c_star <- rbinom(B, n2, p2)
```

```
## success-only continuity correction per replicate where needed
add <- 0.5
add1 <- ifelse(a_star == 0, add, 0)
add2 <- ifelse(c_star == 0, add, 0)

p1_star <- (a_star + add1) / (n1 + 2 * add1) # if a_star==0 -> denom n1+1
p2_star <- (c_star + add2) / (n2 + 2 * add2) # if c_star==0 -> denom n2+1

theta_star <- log(p1_star / p2_star)
theta_star <- theta_star[is.finite(theta_star)] # safety

se_SAFE <- sd(theta_star)
bias_SAFE <- mean(theta_star) - theta_First
theta_SAFE <- theta_First - bias_SAFE # bias-corrected point

## 3) Nicely formatted comparison -----
out_lnRR <- rbind( Point = c(First = theta_First,
  SAFE = theta_SAFE),
  SE = c(First = se_First,
  SAFE = se_SAFE)
)
print(round(out_lnRR, 4))
```

We obtain the following:

```
      First    SAFE
Point -1.6094 -1.4571
SE      0.7135  0.7277
```

The bootstrap SE is slightly larger, and the BC point is pulled closer to zero. Our simulation suggests that the SAFE procedure tends to give more accurate estimates. Also, it is noteworthy that for both lnOR and lnRR, we could have used a multivariate normal to sample using modelling risks: p_1 and p_2 and their sampling variances $p_1(1 - p_1)/n_1$ and $p_2(1 - p_2)/n_2$ (for an example, see an additional material: <https://ejlundgren.github.io/SAFE/>).

5. Less common effect sizes

In this section, we apply the SAFE bootstrap to two lesser-known effect-size statistics to demonstrate its broad applicability.

5.1. Log coefficient-of-variation ratio (lnCVR)

The coefficient of variation (CV) is defined as the ratio of the standard deviation to the mean, $CV = s/\bar{x}$. When comparing two groups in terms of variability (CV), the log CV ratio is useful, as proposed by Nakagawa et al.⁸:

$$\ln \text{CVR}_1 = \ln \left(\frac{s_1/\bar{x}_1}{s_2/\bar{x}_2} \right) = \ln(s_1) - \ln(s_2) - [\ln(\bar{x}_1) - \ln(\bar{x}_2)]. \quad (24)$$

A first-order delta-method approximation for its variance:

$$\text{Var}[\ln \text{CVR}_1] = \frac{s_1^2}{n_1 \bar{x}_1^2} + \frac{s_2^2}{n_2 \bar{x}_2^2} + \frac{1}{2(n_1 - 1)} + \frac{1}{2(n_2 - 1)}. \quad (25)$$

This accounts for both the uncertainty in the means and the sample standard deviations.

Subsequently, Senior et al.⁹ proposed second-order versions of both point and variance estimates as follows:

$$\ln \text{CVR}_2 = \ln \left(\frac{s_1/\bar{x}_1}{s_2/\bar{x}_2} \right) + \frac{1}{2} \left(\frac{1}{n_1 - 1} - \frac{1}{n_2 - 1} \right) + \frac{1}{2} \left(\frac{s_2^2}{n_2 \bar{x}_2^2} - \frac{s_1^2}{n_1 \bar{x}_1^2} \right), \quad (26)$$

$$\begin{aligned} \text{Var}[\ln \text{CVR}_2] &= \frac{s_1^2}{n_1 \bar{X}_1^2} + \frac{s_1^4}{2 n_1^2 \bar{X}_1^4} + \frac{n_1}{2(n_1 - 1)^2} \\ &+ \frac{s_2^2}{n_2 \bar{X}_2^2} + \frac{s_2^4}{2 n_2^2 \bar{X}_2^4} + \frac{n_2}{2(n_2 - 1)^2}. \end{aligned} \quad (27)$$

They also proposed dependent cases, both the first- and second-order versions of sampling variances (D1 and D2, respectively), as follows:

$$\text{Var}[\ln \text{CVR}_{\text{D1}}] = \frac{s_1^2}{n \bar{x}_1^2} + \frac{s_2^2}{n \bar{x}_2^2} - \frac{2r s_1 s_2}{n \bar{x}_1 \bar{x}_2} + \frac{1}{n - 1} - \frac{r^2}{n - 1} \quad (28)$$

$$\text{Var}[\ln \text{CVR}_{\text{D2}}] = \frac{s_1^2}{n \bar{x}_1^2} + \frac{s_1^4}{2 n^2 \bar{x}_1^4} + \frac{s_2^2}{n \bar{x}_2^2} + \frac{s_2^4}{2 n^2 \bar{x}_2^4} - \frac{2r s_1 s_2}{n \bar{x}_1 \bar{x}_2} - \frac{r^2 s_1^2 s_2^2}{n^2 \bar{x}_1^2 \bar{x}_2^2} + \frac{n}{(n - 1)^2} - \frac{r^2}{n - 1} - \frac{r^4}{(n - 1)^2}. \quad (29)$$

SAFE does not require these complex formulas; we can use the quartet with the covariances between the two means and between the two variances, where r is 0 for independent groups and $r > 0$ for dependent cases (paired or matched-case designs):

$$\begin{bmatrix} \bar{x}_1^* \\ \bar{x}_2^* \\ s_1^{2*} \\ s_2^{2*} \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \\ s_1^2 \\ s_2^2 \end{bmatrix}, \underbrace{\begin{bmatrix} \frac{s_1^2}{n_1} & \frac{r s_1 s_2}{\sqrt{n_1 n_2}} & 0 & 0 \\ \frac{r s_1 s_2}{\sqrt{n_1 n_2}} & \frac{s_2^2}{n_2} & 0 & 0 \\ 0 & 0 & \frac{2 s_1^4}{n_1 - 1} & \frac{2 r^2 s_1^2 s_2^2}{\sqrt{(n_1 - 1)(n_2 - 1)}} \\ 0 & 0 & \frac{2 r^2 s_1^2 s_2^2}{\sqrt{(n_1 - 1)(n_2 - 1)}} & \frac{2 s_2^4}{n_2 - 1} \end{bmatrix}}_{\Sigma} \right).$$

As with SMD, this implementation is conditional on a Gaussian working model for the sample means and sample variances; therefore, negative variance draws or strongly skewed positive outcomes indicate that a different sampling model may be preferable.

For each of B draws, we compute

$$\ln \text{CVR}^* = \ln(\sqrt{s_1^{2*}/\bar{x}_1^*}) - \ln(\sqrt{s_2^{2*}/\bar{x}_2^*}).$$

The SAFE summaries are

$$\text{SE}_{\text{SAFE}} = \text{sd}(\ln \text{CVR}^*), \quad \ln \text{CVR}_{\text{BC}} = 2 \ln \text{CVR}_{\text{plug-in}} - \overline{\ln \text{CVR}^*}.$$

Below are two examples, one assuming $r = 0$ (two independent groups) and the other $r = 0.5$ (two paired groups with a correlation of 0.5). In this instance, we create an R function that handles the independent and dependent scenarios more flexibly.

```
set.seed(1234)
library(MASS)

safe_lncvr <- function(x1, s1, n1, x2, s2, n2, r = 0, B = 1e6) {
  # --- Point estimates -----
  lnCVR1 <- log(s1 / x1) - log(s2 / x2) # Eq. 24
  lnCVR2 <- lnCVR1 +
    0.5 * (1/(n1 - 1) - 1/(n2 - 1)) +
    0.5 * (s2^2/(n2 * x2^2) - s1^2/(n1 * x1^2)) # Eq. 26

  # --- Variance formulas (incorporating r) -----
  if (r == 0) {
    # independent groups
    var1 <- s1^2/(n1 * x1^2) + s2^2/(n2 * x2^2) +
      1/(2*(n1 - 1)) + 1/(2*(n2 - 1)) # Eq. 25
    var2 <- s1^2/(n1 * x1^2) + s1^4/(2 * n1^2 * x1^4) + n1/(2*(n1 - 1)^2) +
      s2^2/(n2 * x2^2) + s2^4/(2 * n2^2 * x2^4) + n2/(2*(n2 - 1)^2) # Eq. 27
  } else {
    # dependent (paired) case, assume n1 = n2 = n
    n <- n1
    var1 <- s1^2/(n * x1^2) + s2^2/(n * x2^2) -
      2*r*s1*s2/(n * x1 * x2) +
      1/(n - 1) - r^2/(n - 1) # Eq. 28
    var2 <- s1^2/(n * x1^2) + s1^4/(2 * n^2 * x1^4) +
      s2^2/(n * x2^2) + s2^4/(2 * n^2 * x2^4) -
      2*r*s1*s2/(n * x1 * x2) +
      r^2 * s1^2 * s2^2 * (x1^4 + x2^4) / (2 * n^2 * x1^4 * x2^4) +
      n/(n - 1)^2 - r^2/(n - 1) +
      r^4 * (s1^8 + s2^8) / (2 * (n - 1)^2 * s1^4 * s2^4) # Eq. 29
  }
  se1 <- sqrt(var1)
  se2 <- sqrt(var2)

  # --- SAFE bootstrap for bias correction -----
  mu <- c(x1, x2, s1^2, s2^2)
  Sigma <- matrix(0, 4, 4)
  Sigma[1,1] <- s1^2 / n1
  Sigma[2,2] <- s2^2 / n2
  Sigma[1,2] <- Sigma[2,1] <- r * s1 * s2 / sqrt(n1 * n2)
  Sigma[3,3] <- 2 * s1^4 / (n1 - 1)
  Sigma[4,4] <- 2 * s2^4 / (n2 - 1)
  Sigma[3,4] <- Sigma[4,3] <- 2 * r^2 * s1^2 * s2^2 / sqrt((n1 - 1)*(n2 - 1))

  draws <- MASS::mvrnorm(B, mu, Sigma)
  draws <- draws[draws[,3] > 0 & draws[,4] > 0, ]
  X1s <- draws[,1]; X2s <- draws[,2]
  S1s <- sqrt(draws[,3]); S2s <- sqrt(draws[,4])
  theta_star <- log(S1s / X1s) - log(S2s / X2s)
}
```

```

se_safe <- sd(theta_star)
bias    <- mean(theta_star) - lnCVR1
lnCVR_bc <- lnCVR1 - bias

c(First      = lnCVR1,
   Second    = lnCVR2,
   SAFE_BC   = lnCVR_bc,
   SE_First   = se1,
   SE_Second = se2,
   SE_SAFE    = se_safe)
}

# --- Examples -----

# Independent groups (r = 0) n can be different between two groups
res_ind <- safe_lncvr(x1 = 17, s1 = 2, n1 = 23,
                     x2 = 12, s2 = 3, n2 = 27,
                     r = 0)

# Paired design (r = 0.5; n1 = n2 = 25) n needs to be the same
res_dep <- safe_lncvr(x1 = 15, s1 = 2, n1 = 25,
                     x2 = 10, s2 = 2, n2 = 25,
                     r = 0.5)

# Organise into tables -----
tab_ind <- rbind( Point = res_ind[c("First", "Second", "SAFE")],
                  SE    = res_ind[c("SE_First", "SE_Second", "SE_SAFE")])
)
tab_dep <- rbind( Point = res_dep[c("First", "Second", "SAFE")],
                  SE    = res_dep[c("SE_First", "SE_Second", "SE_SAFE")])
)

```

We should get the following (first, the independent case, and then the dependent case):

```

> round(tab_ind, 4)
      First Second  SAFE
Point -0.7538 -0.7494 -0.7479
SE     0.2118  0.2160  0.2453
> round(tab_dep, 4)
      First Second  SAFE
Point -0.4055 -0.4050 -0.4052
SE     0.1803  0.1853  0.2112

```

This appears to be the most complex script so far, but the principles and SAFE steps remain the same. As above, the second-order delta method yields point estimates similar to those from SAFE.

5.2. Hardy–Weinberg disequilibrium

In population genetics, the Hardy–Weinberg equilibrium (HWE) at a bi-allelic locus implies that observed genotype counts n_{AA}, n_{Aa}, n_{aa} in a sample of size $n = n_{AA} + n_{Aa} + n_{aa}$ follow proportions $p_1 = n_{AA}/n$, $p_2 = n_{Aa}/n$, $p_3 = n_{aa}/n$ with $p_1 + p_2 + p_3 = 1$. The allele frequency of A is

$$p = \frac{2n_{AA} + n_{Aa}}{2n} = p_1 + \frac{1}{2}p_2,$$

so the expected heterozygote proportion under HWE is $2p(1 - p)$.

We quantify departure from HWE (HWD) using the relative excess of heterozygosity index as a measure of HWD,¹⁰ which compares the observed heterozygosity to a symmetric homozygote-based benchmark:

$$\ln \omega = \ln \left(\frac{p_2}{2 \sqrt{p_1 p_3}} \right). \quad (30)$$

This equals 0 under exact HWE (since $2\sqrt{p_1 p_3} = 2p(1-p)$ at HWE) and is label-invariant with respect to *AA* and *aa*.

A first-order delta-method approximation for the sampling variance of $\ln \omega$ is

$$\text{Var}[\ln \omega] = \frac{1}{n} \left(\frac{1}{p_2} + \frac{1-p_2}{4 p_1 p_3} \right). \quad (31)$$

As you can see, these equations (Eqs. (30) and (31)) are undefined when $p_1 = 0$, $p_2 = 0$, and $p_3 = 0$, and as $\ln \text{OR}$ and $\ln \text{RR}$, we can add 0.5 in such cases (i.e., continuity correction).

To generate SAFE estimates of the departure from HWE, we model genotype counts as a single draw from a multinomial distribution:

$$(n_{AA}^*, n_{Aa}^*, n_{aa}^*) \sim \text{Multinomial}(n; p_1, p_2, p_3).$$

Then, for each of B replicates, compute

$$p_{AA}^* = n_{AA}^*/n, \quad p_{Aa}^* = n_{Aa}^*/n, \quad p_{aa}^* = n_{aa}^*/n, \quad \ln \omega^* = \ln \left(\frac{p_{Aa}^*}{2 \sqrt{p_{AA}^* p_{aa}^*}} \right).$$

The SE for the effect size can then be found as

$$\text{SE}_{\text{SAFE}} = \text{sd}(\ln \omega^*), \quad \ln \omega_{\text{BC}} = 2 \ln \omega - \overline{\ln \omega^*}.$$

Below is an example implemented in R:

```
set.seed(123)

## --- observed data -----
nAA <- 40; nAa <- 25; naa <- 50
n    <- nAA + nAa + naa

## --- continuity correction (CC) settings -----
add <- 0.5

## --- apply CC to observed counts only if any zero -----
add_obs <- if (any(c(nAA, nAa, naa) == 0)) add else 0
n_eff_obs <- n + 3 * add_obs      # no. of genotypes = 3
p1_hat <- (nAA + add_obs) / n_eff_obs
p2_hat <- (nAa + add_obs) / n_eff_obs
p3_hat <- (naa + add_obs) / n_eff_obs

## --- plug-in point & delta-method SE (use guarded p-hats) -----
omega_hat <- log(p2_hat / (2 * sqrt(p1_hat * p3_hat))) # Eq. 30
t2 <- (1 - p2_hat) / (4 * p1_hat * p3_hat) + 1 / p2_hat
se_delta <- sqrt(t2 / n) # keep true n for delta SE # Eq. 31

## --- SAFE bootstrap (Multinomial) -----
B <- 1e6
draws <- rmultinom(B, n, c(p1_hat, p2_hat, p3_hat)) # multinomial
```

```
## Per-replicate CC if any simulated category is zero
needs_cc <- colSums(draws == 0) > 0
if (any(needs_cc)) {
  draws[, needs_cc] <- draws[, needs_cc] + add # add 0.5 to all three categories
}
denom <- rep(n, B)
denom[needs_cc] <- n + 3 * add # 3 = the number of genotype categories

pAA_star <- draws[1, ] / denom
pAa_star <- draws[2, ] / denom
paa_star <- draws[3, ] / denom

omegas <- log(pAa_star / (2 * sqrt(pAA_star * paa_star)))

## --- SAFE summaries -----
se_SAFE <- sd(omegas)
bias_hat <- mean(omegas) - omega_hat
omega_BC <- omega_hat - bias_hat

## --- summary table -----
res <- rbind( Point = c(First = omega_hat, SAFE = omega_BC),
             SE    = c(First = se_delta, SAFE = se_SAFE)
)
round(res, 4)
```

From this script, we get the following:

	First	SAFE
Point	-1.2747	-1.2654
SE	0.2264	0.2319

Our simulation shows that the SAFE procedure yields a less biased point estimate when the sample size is small, but it can also overestimate the SE when the sample size is small (see our additional material: <https://ejlundgren.github.io/SAFE/>).

6. Applying SAFE procedures

Table 1 summarises the settings for the effect sizes above, the summary statistics required, the working sampling models used, and the main caveats. In all cases, SAFE proceeds by specifying a working model for the reported summary statistics, drawing bootstrap replicates under that model, transforming each replicate into the effect size of interest, and then summarising the resulting bootstrap cloud. What differs across designs is the choice of sampling model and the assumptions needed to justify it. Notably, the SAFE implementations presented above are most appropriate for effect sizes that can be expressed on an approximately unbounded, reasonably symmetric scale and parameterised from commonly reported descriptive statistics; SAFE should not be applied to highly skewed or bounded raw effect sizes without adopting a more appropriate sampling model.

Important, so far, every example has focused on computing an effect size for single comparison, chiefly to make the four SAFE steps clear. In reality, however, meta-analyses, comprise multiple pairwise comparisons (e.g., means, SD, and n). Fortunately, our procedure parallelises naturally. In R for example, we can wrap the SAFE function in `mapply()` or `pmap()` from the `purrr` package,³¹ which dispatches the function row-wise and binds the results in to a single data frame (parallelised functions are available for the effect sizes introduced above as part of an additional material: <https://ejlundgren.github.io/SAFE/>).

The R script below illustrates the workflow for log response ratios (ratio of means; `safe_RoM`) using a meta-analytic dataset from Curtis and Wang,³² which is a part of the `metadat` package.³³ As shown

Table 1. Practical notes for applying SAFE to the effect sizes covered in this tutorial.

Design/effect size	Summary statistics required	Working sampling model in SAFE ('fit')	Key assumptions/caveats
Independent-group lnRoM (ratio of means)	$\bar{x}_1, s_1, n_1, \bar{x}_2, s_2, n_2$	Bivariate Gaussian model for the two sample means	It requires positive means for the log transform; a multivariate-normal working model could be used if negative values are rare.
Independent-group SMD (standardised mean difference)	$\bar{x}_1, s_1, n_1, \bar{x}_2, s_2, n_2$	Joint model for $(\bar{x}_1, \bar{x}_2, s_1^2, s_2^2)$, here approximated by a multivariate Gaussian working model	Mean–variance independence assumed; Gaussian variance sampling can produce impossible negative draws, so χ^2 /Wishart-based alternatives may be preferable.
Independent-group lnCVR (coefficients of variation ratio)	$\bar{x}_1, s_1, n_1, \bar{x}_2, s_2, n_2$	Joint model for $(\bar{x}_1, \bar{x}_2, s_1^2, s_2^2)$, again typically approximated by a multivariate Gaussian working model	It requires positive means and variances; inherits the same caveats as SMD regarding Gaussian approximation for sample variances.
Paired/repeated-measures lnRoM, SMD, or lnCVR	Same as above, plus within-pair correlation r (or a plausible assumed value)	Multivariate Gaussian model with covariance terms determined by r , including covariance between means and between sample variances	When r is not reported, analysts may use a plausible assumed value (e.g., $r = 0.5$), but sensitivity analyses over a range of plausible r values should be conducted and reported.
lnOR (odds ratio)/lnRR (risk ratio) from 2×2 tables	a, b, c, d or equivalently events and totals in each group	Independent binomial models for event counts in each group (equivalently, two-category multinomial models with fixed row totals)	Zero cells require continuity correction; the resulting SAFE estimates are then for the corrected estimator.
Hardy–Weinberg disequilibrium	Genotype counts (n_{AA}, n_{Aa}, n_{aa})	Multinomial model for the genotype-count vector	Zero categories require continuity correction; the resulting SAFE estimates are then for the corrected estimator.

Note: The table summarises the required reported summaries, the working sampling model used in the SAFE 'fit' step, and the key assumptions or caveats for each setting.

below, one instance of `mapply()` returns two named columns (`y_SAFE` and `v_SAFE`), respectively, containing the SAFE point estimates and variances for each pairwise set of summary statistics. Note that, for comparison, we also have point estimates and variances (`yi`, `vi`) obtained via `escalc()` from the `metafor` package.²²

```
## ---- 0.  preparations -----
library(metafor)
library(metadat)
set.seed(777)

## ---- 1.  SAFE log response ratios -----
# dataset and sampling data row indices
dat1 <- dat.curtis1998[sample(nrow(dat.curtis1998), 10), c(1, 15:20)]

# functions for lnRoM
safe_RoM <- function(x1, x2, s1, s2, n1, n2, B = 1e6) {
  # plug-in estimate
  t0 <- log(x1/x2)
  # draws
  x1_star <- rnorm(B, mean = x1, sd = s1/sqrt(n1))
  x2_star <- rnorm(B, mean = x2, sd = s2/sqrt(n2))
  # transform, drop any non-positive draws (if any)
  ts <- log(x1_star[x1_star>0 & x2_star>0] / x2_star[x1_star>0 & x2_star>0])
  # bias, bias-corrected point estimate and variance
  c(y_SAFE = 2*t0 - mean(ts),
    v_SAFE = var(ts)
  )
}

dat1 <- escalc(measure = "ROM",
              m1i = dat1$m1i, sd1i = dat1$sd1i, n1i = dat1$n1i,
              m2i = dat1$m2i, sd2i = dat1$sd2i, n2i = dat1$n2i,
              data = dat1, append = TRUE)

safe_out1 <- mapply( safe_RoM,
                    x1 = dat1$m1, x2 = dat1$m2,          # means
                    s1 = dat1$sd1, s2 = dat1$sd2,        # SDs
                    n1 = dat1$n1, n2 = dat1$n2,          # sample sizes
                    SIMPLIFY = TRUE)

# round for easy comparisons, but usually not recommended
dat1$yi_SAFE <- round(safe_out1["y_SAFE", ], 4)
dat1$vi_SAFE <- round(safe_out1["v_SAFE", ], 4)
```

If you run this script, we will get the following:

```
> dat1
```

	id	m1i	sd1i	n1i	m2i	sd2i	n2i	yi	vi	yi_SAFE	vi_SAFE
24	242	2.2000	0.411100	10	1.330	0.189700	10	0.5033	0.0055	0.5041	0.0056
1	21	6.8169	1.769982	3	3.945	1.115797	5	0.5470	0.0385	0.5503	0.0406
90	739	16.5900	2.914900	6	13.800	1.592200	6	0.1841	0.0074	0.1856	0.0074
60	456	38.3600	3.215000	2	26.490	1.790000	2	0.3702	0.0058	0.3708	0.0058
62	458	186.2600	11.834000	2	180.480	11.867000	2	0.0315	0.0042	0.0314	0.0042
83	726	16.9800	1.959600	6	10.960	1.935100	6	0.4378	0.0074	0.4362	0.0075
12	96	72.5000	15.000000	24	60.500	14.000000	24	0.1809	0.0040	0.1807	0.0040
10	87	71.7200	14.300000	16	59.940	14.280000	16	0.1794	0.0060	0.1788	0.0061
76	615	374.4000	109.567300	5	298.200	96.374500	5	0.2276	0.0380	0.2258	0.0400
25	254	23.1100	0.844300	22	14.940	0.844300	22	0.4362	0.0002	0.4362	0.0002

The same process can be implemented using the log risk ratio (lnRR; safe_RR), with the dataset from Graves et al.,³⁴ which is also part of the metdat package.

```
## ---- 2. SAFE log risk ratios -----
# dataset and sampling data row indices (note should be dat.graves1996)
dat2 <- dat.graves2010[sample(nrow(dat.graves2010), 10), ]

# function for lnRR -----
# note the function is slightly different from the above
# (it takes n1 and n2 rather than b and d
safe_RR <- function(a, c, n1, n2, B = 1e6,
                    add = 0.5) { # continuity correction

  ## continuity correction -----
  add1 <- if (a == 0) add else 0          # only if the observed cell is zero
  add2 <- if (c == 0) add else 0
  # note the +add on *both* successes & failures
  p1_hat <- (a + add1) / (n1 + add1*2)
  p2_hat <- (c + add2) / (n2 + add2*2)
  lnRR_raw <- log(p1_hat / p2_hat)
  # draw once
  a_star <- rbinom(B, n1, a / n1)
  c_star <- rbinom(B, n2, c / n2)
  # transform
  add1_star <- ifelse(a_star == 0, add, 0)
  add2_star <- ifelse(c_star == 0, add, 0)
  p1_star <- (a_star + add1_star) / (n1 + add1_star*2)
  p2_star <- (c_star + add2_star) / (n2 + add2_star*2)
  lnRR_star <- log(p1_star / p2_star)
  # bias, bias-corrected point estimate and variance
  c(y_SAFE = 2*lnRR_raw - mean(lnRR_star),
    v_SAFE = var(lnRR_star)
  )
}

dat2 <- escalc(measure = "RR",
              ai = dat2$ai, ci = dat2$ci,          # events in ctrl/treat
              n1i = dat2$n1i, n2i = dat2$n2i,
              data = dat2,
              add = 0.5, to = "only0", append = TRUE)

safe_out2 <- mapply( safe_RR,
  a = dat2$ai, n1 = dat2$n1,          # events + totals (grp1)
  c = dat2$ci, n2 = dat2$n2,          # events + totals (grp2)
  SIMPLIFY = TRUE)

# round for easy comparisons, but usually not recommended
dat2$yi_SAFE <- round(safe_out2["y_SAFE", ], 4)
dat2$vi_SAFE <- round(safe_out2["v_SAFE", ], 4)
```

We should now get this data frame.

```
> dat2
```

	study	ai	n1i	ci	n2i	yi	vi	yi_SAFE	vi_SAFE
12	Azurin 1965-ii	92	148100	52	48933	-0.5369	0.0301	-0.5410	0.0308
16	Saroso 1978-i	18	156300	19	79250	-0.7332	0.1082	-0.7316	0.1186
9	PCC 1973a-iii	36	44450	23	11050	-0.9439	0.0711	-0.9527	0.0756
8	PCC 1973a-ii	33	45750	22	11050	-1.0153	0.0756	-1.0233	0.0806
3	Mosley 1970-i	32	11491	13	3810	-0.2031	0.1078	-0.2286	0.1205
17	Saroso 1978-ii	10	155600	18	79250	-1.2625	0.1555	-1.2367	0.1819
1	Oseasohn 1965	8	6956	33	7103	-1.3962	0.1550	-1.3413	0.1911

15	PCC	1968	54	268700	41	90900	-0.8084	0.0429	-0.8116	0.0443
10	PCC	1973a-iv	26	44700	22	11050	-1.2305	0.0838	-1.2341	0.0896
13	Azurin	1965-iii	71	143600	52	48933	-0.7651	0.0333	-0.7679	0.0342

As you can see, SAFE provides very similar point estimates and variance estimates as `escalc()`, though the variance estimates are often larger (more conservative), and this discrepancy disappears as the sample size grows. In practice, for many existing studies, SAFE may not change the qualitative conclusions. However, the SAFE procedure future-proofs the analysis for new effect-size statistics. Additionally, one can compare the point estimate and variance derived from the delta method for new effect size statistics with those obtained using the SAFE bootstrap, as we have done in this tutorial.

7. Concluding remarks

In this tutorial, we have introduced the SAFE bootstrap that builds on previous single-fit parametric methods.^{6,7} SAFE bootstrap provides a practical and transparent model-based route to bias correction and SE estimation when only summary statistics are available. By following the four-step recipe (fit, draw, transform, and summarise), researchers can avoid deriving a new delta-method formula for every new effect size, while making the assumed sampling model explicit. This approach lowers the barrier to obtaining model-based BC point estimates and SEs for many effect sizes that can be expressed as functions of reported summary statistics, provided an appropriate sampling model can be specified. Although this is not the focus of our article, our simulation validation suggests that in the settings considered here, SAFE point estimates could outperform plug-in and small-sample-corrected formulas, especially when sample sizes are small. In line with our results, the simulation work of Fletcher and Jowett⁷ demonstrated that confidence intervals (CIs) derived from the SAFE procedure (i.e., single-fit bootstrapping) outperformed those obtained using the delta method, suggesting that single-fit bootstrap procedures can outperform delta-method-based CIs in some settings. Yet, we also note that delta-method-based variance estimators could outperform their SAFE counterparts in some scenarios; SAFE sampling variance tends to be higher than that of the delta method (see an additional material: <https://ejlundgren.github.io/SAFE/>).

Here, we have also shown that the SAFE procedure can be applied to familiar effect size statistics, such as $\ln RoM$, SMD , $\ln OR$, and $\ln RR$. Yet the true potential of the SAFE bootstrap lies in its adaptability. The same reasoning and coding pipeline can be applied to less common but valuable metrics, such as $\ln CVR$ ^{8,9} or the HWE statistic,¹⁰ as illustrated. Additionally, for some effect sizes, variance estimators are not derived for paired or dependent designs; however, this is not an issue for the SAFE approach, which can readily handle dependence. This flexibility allows researchers to choose the most suitable effect size for their specific question, rather than being restricted to effect sizes for which estimators of the sampling variance exist.

A natural next step will be to consider whether SAFE can be extended beyond the two-group designs treated here, for example, to sample correlations, R^2 , regression coefficients, or more complex effect sizes arising from longitudinal, covariate-adjusted, or multilevel models. However, these extensions are not straightforward. In many such settings, the primary study already reports SEs for the target estimate, so SAFE would mainly be useful when one wishes to transform the estimate or construct a more complex derived effect size. Relatedly, SAFE might be useful for converting effect size statistics; correlation, $\ln OR$, and SMD are convertible to each other,¹⁸ but these formulas are likely to introduce bias due to their non-linear nature. It is also an interesting avenue for future research to apply the SAFE procedure to the concept of ‘smooth variance estimators’, in which sampling variance is calculated by averaging observed parameter values. For example, this approach can be applied to CV , which is a part of the sampling variance formula for $\ln RoM$, improving the estimation of overall (meta-analytic) means.³⁵

With SAFE, the initial ‘fit’ step is the crucial one, because it requires choosing a sampling model for the reported statistics. In this tutorial, we have made heavy use of Gaussian working models for continuous outcomes and binomial or multinomial models for discrete outcomes, as these are often parameterisable from the summary information reported in primary studies. However, SAFE does not remove the need for statistical judgement: if the working model is poor, the resulting bias correction and SE may also be poor.

In principle, one could instead use positive-support models, such as log-normal or gamma models for some continuous outcomes, but these often require additional assumptions or parameterisation choices that are not identifiable from the usual reported summaries alone.³⁶ Extending SAFE to such settings remains an important direction for future work. Heuristically, the SAFE bootstrap cloud can resemble the way one works with a Bayesian posterior distribution, in the sense that inference is based on a simulated distribution of the quantity of interest rather than a closed-form approximation.³⁷

Notably, traditional bootstrap procedures applied directly to raw data remain an important benchmark.^{1–4} When raw data are available, or if a primary study already reports bootstrap-based effect-size estimates and corresponding SEs or confidence intervals derived from the raw data, those should generally be preferred. SAFE is designed for the more typical meta-analytic situation in which only summary statistics are available. Accordingly, SAFE is most useful when raw data are unavailable and one needs a transparent, model-based way to obtain BC effect-size estimates and SEs from reported summaries, especially when sampling variance formulas are unavailable, difficult to derive, or potentially inaccurate in small samples. However, SAFE is not assumption-free: it should only be used when the analyst can specify a plausible working model for the sampling distribution of the reported statistics.

Ultimately, we believe the SAFE bootstrap offers a useful addition to the meta-analyst's toolkit. We do not propose SAFE as a replacement for elegant analytic formulas or for raw-data bootstrap procedures when those are available. Rather, SAFE fills an important practical gap: it offers a computationally efficient, transparent, and reproducible summary-statistic approach when raw data are unavailable and when analytic variance formulas are unavailable, difficult to derive, or potentially inaccurate in small samples. In that role, SAFE can both support applied work and help researchers build a deeper intuition for sampling distributions, transformation-induced bias, and simulation-based inference.^{38–41} Therefore, the path to more accessible meta-analytic inference is, as we have shown, a SAFE one.

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